Seungbeom Seo

M.S. Student, Biomedical Engineering
Brain–Computer Interface Lab (Prof. Sung-Phil Kim)
Ulsan National Institute of Science and Technology (UNIST)

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RESEARCH INTERESTS

Brain–Computer / Brain–Machine Interfaces · Neural Encoding & Decoding of Vision and Motor Control · Cognitive Neuroscience of Sensory Perception · Brain Imaging

SKILLS

TOOLS	MATLAB · Python · EEGLAB · SPM · FSL · MySQL
METHODS	EEG · fMRI · Eye-tracking

EDUCATION

2025 – Current
M.S. in Biomedical Engineering
Brain–Computer Interface Lab (Advisor: **Prof. Sung-Phil Kim**) · UNIST, Ulsan, South Korea

2019 – 2025
B.S. in Biomedical Engineering
Minor in Electrical Engineering · UNIST, Ulsan, South Korea

2023 – 2024
Republic of Korea Air Force – Tactical Air Control Party
Air Support Operations Center · Mandatory Military Service

2016 – 2018
Hansung Science High School
Seoul, South Korea

RESEARCH EXPERIENCE

2020 – Current
Brain–Computer Interface Lab, UNIST
Advisor: **Prof. Sung-Phil Kim**

- Led human neurophysiology studies using EEG (and EEG + eye-tracking) with end-to-end responsibility from experimental design to analysis (MATLAB / EEGLAB; Python as needed).
- Key contributions across projects: paradigm development and implementation, participant recruitment and session running, preprocessing / feature extraction, and interpretation / reporting of results.

SELECTED PROJECTS

- **EEG responses of emotion across OTT subscription fees** *(joint project with Korea University)* — Designed and implemented the experimental paradigm; conducted EEG experiments with 10+ participants; analyzed EEG data to characterize emotional differences across cost conditions.
- **Neural basis of the choice-overload effect** — Developed and executed a complex EEG + eye-tracking paradigm; conducted experiments with 50+ participants; processed and analyzed large-scale datasets to probe neural processes under different preference conditions.
- **fMRI study on tactile illusion** — Conducted 3T fMRI experiments with 30+ participants; analyzed BOLD signals using MATLAB / SPM / FSL; compared EEG vs. fMRI methodologies for sensory-mechanism studies.
- **EEG signal analysis** — Signal processing and analysis using EEGLAB and MATLAB.

Spring 2022

Decision Neuroscience & Cognitive Engineering Lab, UNIST

Advisor: **Prof. Dongil Chung**

- **Resting-state EEG analysis of drug-induced parkinsonism and Parkinson's disease** *(joint project with Dongtan Sacred Heart Hospital)* — Analyzed resting-state EEG from PD and DIP patients; performed spectral analyses and EEG-clinical correlation analyses to identify differentiating neural signatures and support diagnostic insight.

Summer 2020

Bio and Neural Interfaces Lab, KAIST

Advisor: **Prof. Seongjun Park**

- **Mechanical interaction of thermally drawn fiber implants with brain tissue** — Investigated tissue-device interactions for fiber-based neural implants using in vivo surgery, histology, and computational modeling. Performed mouse-brain implant surgeries with fiber-based neural probes; verified immune responses via immunohistochemistry; conducted finite-element analysis to model probe-tissue mechanics.

PUBLICATIONS

S. Seo, O.-S. Kwon & S.-P. Kim. *Closed-loop brain-computer interface training may bind imagined movement to its visual effect.* IN PREPARATION

2026 **S. Seo**, S. Lee, N. Lee & S.-P. Kim. *Reducing cognitive demand fails to mitigate choice overload: Multimodal neural and gaze evidence for effort conservation.* **iScience**, 29, 117616.

2024 **S. Seo**, S. Kim, S.-P. Kim, J.-H. Kim, S. Y. Kang & D. Chung. *Low-frequency EEG power and coherence differ between drug-induced parkinsonism and Parkinson's disease.* **Clinical Neurophysiology**, 168, 131–138.

2023 J. Park, S. Wang, S. Lee, **S. Seo**, N. Lee & S. Kim. *Viewer emotional response to webtoon-based drama: an EEG analysis.* **International Journal of Human–Computer Interaction**, 1–15.

2021 K. Kim, C. Sung, J. Lee, J. Won, W. Jeon, **S. Seo**, K. Yoon & S. Park. *Computational and histological analyses for investigating mechanical interaction of thermally drawn fiber implants with brain tissue.* **Micromachines**, 12(4), 394.

2021 S. Lee, **S. Seo**, N. Lee & S.-P. Kim. *The effect of OTT subscription fee on content engagement.* **The Journal of Korean Institute of Communications and Information Sciences**, 46(3), 572–582.

CONFERENCE PRESENTATIONS

Nov 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *Closed-loop motor-imagery brain–computer interface training reshapes visual-motion perception.* **Society for Neuroscience (SfN) 2026**, Washington, DC. POSTER

Oct 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *Closed-loop motor-imagery BCI training transfers to perception of the stimulus it controls.* **2026 Korea BCI Society Fall Conference**, Suwon, South Korea. POSTER

Jul 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *EEG neurometric–psychometric dissociation in visual motion perception.* **Brain Engineering Society of Korea 2026**, Jeju, South Korea. TALK

May 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *EEG neurometric function mirrors psychometric performance in visual motion perception.* **Vision Sciences Society (VSS) 2026**, St. Pete Beach, FL. POSTER

Feb 2026 J. Kim, C.-E. Yoon, **S. Seo** & S.-P. Kim. *Real-time single-trial decoding in a calibration-free P300-based BCI using a compact eight-channel EEG configuration.* **14th International Conference on Brain–Computer Interface (BCI) 2026**, Jeongseon, South Korea. POSTER

Sep 2024 **S. Seo**, S. Kim, S.-P. Kim, J. Kim, D. Chung & S. Y. Kang. *Quantitative EEG analysis for comparing drug-induced parkinsonism and Parkinson's disease.* **International Congress of Parkinson's Disease and Movement Disorders 2024**, Philadelphia, PA. POSTER

Nov 2022 J.-H. Kim, J. Kim, **S. Seo** & S.-P. Kim. *Somatosensory cortical representations of assimilation effects by vibrotactile stimulation: an fMRI study.* **Society for Neuroscience (SfN) 2022**, San Diego, CA. POSTER

Sep 2022 **S. Seo**, S. Lee, N. Lee & S.-P. Kim. *Preference modulates the effect of choice overload with less involvement of working memory.* **International Congress of Physiological Anthropology 2022**, Eugene, OR. POSTER

May 2022 S. Lee*, **S. Seo***, N. Lee & S.-P. Kim. *Attenuated working memory accounts for choice overload in highly preferred alternatives.* **25th Annual Meeting of the Korean Society for Brain and Neural Sciences**, Incheon, South Korea. POSTER

** Equal contribution.*

RESEARCH GRANTS

2026.09 – Current

AX(AI Transformation)-Leading Entrepreneurial Talent Development Program

Entrepreneurial Member

Ministry of Science and ICT (MSIT) · 2026.09 – 2027.08

2026.07 – Current

Research Center for AI-Convergent Consciousness Studies

Graduate Researcher

Ministry of Science and ICT (MSIT) · 2026.07 – 2030.12

2026.03 – Current

Augmentation Technology for Overcoming the Limits and Disorders of Human Cognition and Intelligence

Graduate Researcher

Information Technology and Telecommunications Planning & Evaluation Institute (IITP) · 2024.07 – 2031.12

2025.09 – Current

Music BrainCoder: A Non-Invasive BCI for Reproducing Imagined Music and Enhancing Musical Performance

Graduate Researcher

National Research Foundation of Korea (NRF) · 2025.03 – 2030.02

2022.08 – 2023.02

Future Brain Convergence Technology Development Project

Undergraduate Researcher

"Development of brain-inspired computational models of active and adaptive intelligence for AI applications."
Ministry of Science and ICT (MSIT) · 2019.06 – 2023.12

2022.04 – 2023.02

Brain Science Proprietary Technology Development Project

Undergraduate Researcher

"Development of analysis methods for ultra-high-resolution neuroimaging signals and information-processing models of neural networks." Ministry of Science and ICT (MSIT) · 2019.06 – 2023.12

2022.03 – 2022.07

Smart Healthcare Research Center

Undergraduate Researcher

2020.07 – 2022.02

Smart Media Services Innovation Convergence Project

Undergraduate Researcher

"Development of neuro-artificial-intelligence convergence technology for next-generation OTT service."
Ministry of Science and ICT (MSIT) · 2020.07 – 2027.12

HONORS & AWARDS

2026 **Outstanding Paper Award**
Brain Engineering Society of Korea 2026 (oral presentation) · Jeju, South Korea

2022 **Excellent Research Award, 2nd Place**
Department of Biomedical Engineering Faculty, UNIST, South Korea

2021 **Student Creative Research Project, 1st Place**
Smart Media Service Research Center, Korea University, South Korea

2020 **Outstanding Grades for the Semester**
Article 62 of the School Regulations · UNIST, South Korea

2020 **Student Creative Research Project, 2nd Place**
Smart Media Service Research Center, Korea University, South Korea

2019 **Outstanding Grades for the Semester**
Article 62 of the School Regulations · UNIST, South Korea

2019 – Current **National Science & Engineering Undergraduate Scholarship**
Korea Student Aid Foundation (Korean Government) · **Full scholarship**

TEACHING EXPERIENCE

Fall 2026 **AI-based Neural Data Science**
Teaching Assistant · UNIST

Spring 2026 **Machine Learning Methods for Neuroengineering**
Teaching Assistant · UNIST

Fall 2025 **Brain & Cognitive Engineering**
Teaching Assistant · UNIST

SERVICES & OTHER PROFESSIONAL ACTIVITIES

PEER REVIEW

2026 **Journal of Neural Engineering**
Co-reviewer (with **Prof. Sung-Phil Kim**) · IOP Publishing

2026 **Frontiers in Network Physiology**
Reviewer

2026 **Frontiers in Digital Health**
Reviewer

UNIVERSITY & DEPARTMENTAL SERVICE

2022 **Undergraduate Student Council, UNIST**
Election Campaign · Emergency Response Committee

2021 **College of Information & Biotechnology Student Council, UNIST**
Division of Academic Studies, School of Electrical Engineering

2020 **School of Biomedical Engineering Student Council, UNIST**
Preparation Task Force

2020 **School of Electrical & Computer Science Engineering Student Council, UNIST**
Emergency Response Committee · Planning Department

2020 **Campus Assistant, Freshman Support Center, UNIST**
Mentor for 18 first-year students (1 year)

2020 **School of Human Factors Engineering Student Council, UNIST**
Planning Department